Math Kangaroo Practice

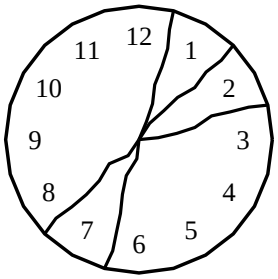
Grade 3-4 · 20 questions · harder · seed 92270

Part A: Each question is worth 3 points

1. How many digits are needed to write down all the numbers from 1 to 300?

A. 792 ***B.*** 790 ***C.*** 789 ***D.*** 600 ***E.*** 300

2. The face of a clock is cracked into 5 pieces as shown.

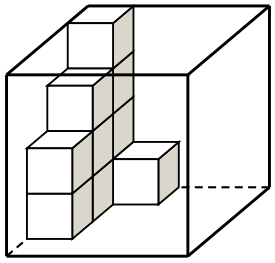


Which piece has the largest sum of numbers?

A. the piece with the 2 ***B.*** the piece with the 7 ***C.*** the piece with the 6

D. the piece with the 1 ***E.*** the piece with the 12

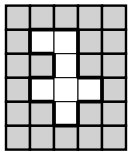
3. Leo has a glass aquarium shaped like a cube, 4 dm long, 4 dm wide and 4 dm tall. Leo has started to put cubes with edges of 1 dm into it, as in the picture. Every cube stands on the bottom or on another cube.

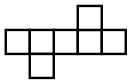
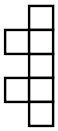
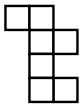
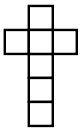
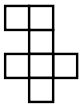


At most how many more such cubes can Leo put into the aquarium?

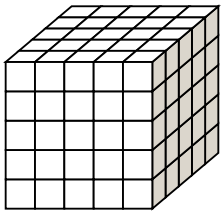
A. 64 ***B.*** 59 ***C.*** 55 ***D.*** 54 ***E.*** 53

4. Which piece covers the empty area exactly? The pieces may be turned and flipped over.



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

5. A big cube made of $5 \times 5 \times 5 = 125$ small cubes is painted on the outside, also underneath, and then taken apart.

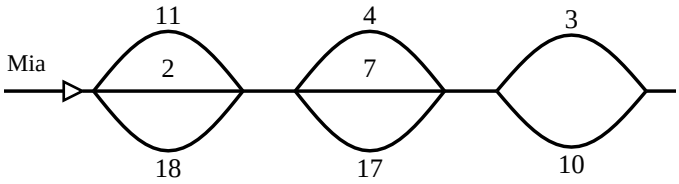


How many small cubes have exactly 3 painted faces?

A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

Part B: Each question is worth 4 points

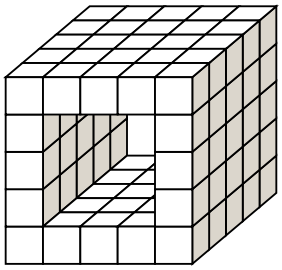
6. Mia walks along the road from left to right and adds up the numbers along the way.



Which total can Mia *not* get?

A. 12 ***B.*** 21 ***C.*** 24 ***D.*** 31 ***E.*** 35

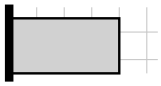
7. Nina built this tunnel out of identical cubes. The hole goes straight through from the front to the back. Nina wants to fill the hole completely, so that the tunnel becomes one solid block.



How many more cubes are needed?

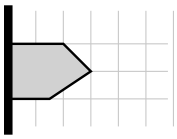
A. 9 ***B.*** 36 ***C.*** 45 ***D.*** 80 ***E.*** 89

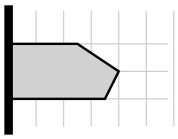
8. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Tom draws the shadow at five different moments. Spread out flat, the flag is 4 squares wide and 2 squares tall (see the picture).

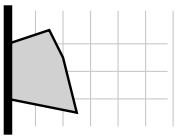


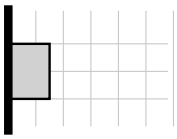
the flag, flat

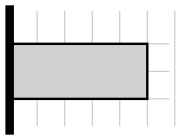
Which shadow cannot be right?

A. 

B. 

C. 

D. 

E. 

9. Dan has a rope 4 m long and cuts it into pieces that are each 40 cm long.
How many pieces does Dan get?

A. 11 ***B.*** 10 ***C.*** 9 ***D.*** 8 ***E.*** 1

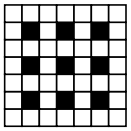
10. Lily has three cards with different digits. Two of them are 1 and 8; the third is face down. Using each card exactly once, with +, –, ×, ÷ and brackets, Lily gets 25.
Which of these digits can be on the hidden card?

A. 9 ***B.*** 7 ***C.*** 5 ***D.*** 4 ***E.*** 3

11. The diagrams below are the 1st, 2nd and 3rd of a pattern. How many black cells will the 6th diagram have?

8 white cells

21 white cells


40 white cells

A. 36 ***B.*** 37 ***C.*** 40 ***D.*** 45 ***E.*** 49

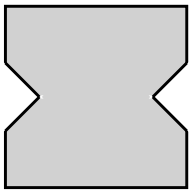
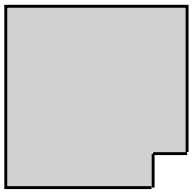
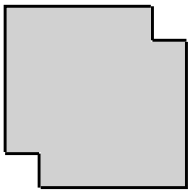
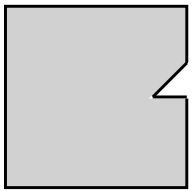
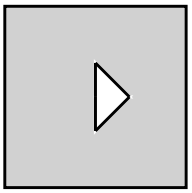
12. Sofia, Zoe, Nicolas, Chloe and Lucas each play a different instrument: the harp, the flute, the xylophone, the drum and the violin. Chloe doesn't play the drum. Lucas uses a bow. Zoe never uses sticks. Nicolas plays an instrument with strings.
Which statement is true?

A. Sofia plays the flute ***B.*** Sofia plays the drum ***C.*** Sofia plays the harp
D. Chloe plays the violin ***E.*** Chloe plays the flute

13. A counter shows 13543. Every second it goes up by 1.
How many seconds until it shows a number whose 5 digits are all the same?

A. 8680 ***B.*** 8679 ***C.*** 8678 ***D.*** 2432 ***E.*** 8

14. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). One corner of the small square is cut off, and the paper is unfolded.
Which design *can* be made this way?

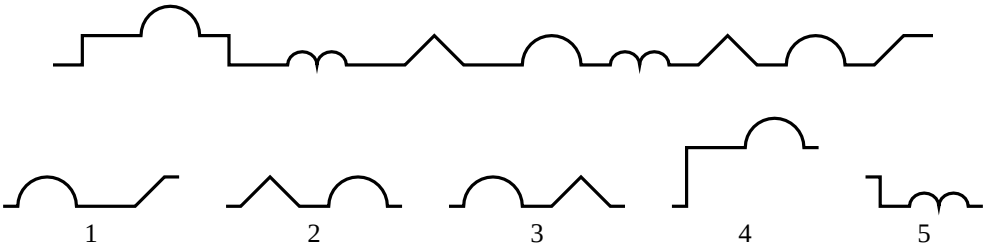
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part C: Each question is worth 5 points

15. Exactly 531 digits are used to write all the house numbers in a street, starting from 1.
How many houses are there?

A. 114 ***B.*** 123 ***C.*** 177 ***D.*** 212 ***E.*** 213

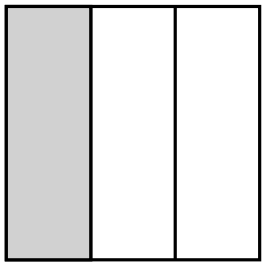
16. This is the outline of a town seen from the river at sunset.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

17. Mia cuts a square photo into 3 equal strips as shown. The perimeter of each strip is 40 cm.



How long is the side of the square?

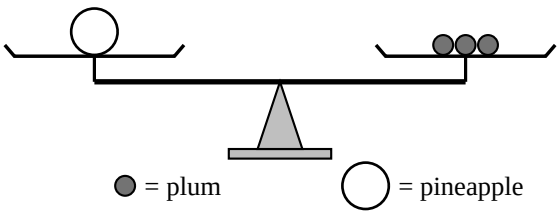
A. 25 cm ***B.*** 15 cm ***C.*** 14 cm ***D.*** 10 cm ***E.*** 5 cm

18. A bar code is made of 22 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 4 more white bars than wide black bars.

How many narrow black bars are there?

A. 7 ***B.*** 6 ***C.*** 5 ***D.*** 4 ***E.*** 3

19. The balance shows that one pineapple weighs as much as 3 plums. 1 pineapple and 3 plums weigh 450 g altogether.



How much does one plum weigh?

A. 25 g ***B.*** 75 g ***C.*** 125 g ***D.*** 150 g ***E.*** 225 g

20. Basil and Peter are twins. Basil was born at 11:47 p.m. on December 31, 2016. Peter was born 45 minutes later.

What is Peter's date of birth?

A. December 31, 2016 ***B.*** January 1, 2017 ***C.*** January 1, 2015

D. December 31, 2017 ***E.*** January 2, 2017

Answer Key

1.	**A**	**2.**	**E**	**3.**	**D**	**4.**	**E**	**5.**	**E**	**6.**	**C**
7.	**C**	**8.**	**E**	**9.**	**B**	**10.**	**E**	**11.**	**A**	**12.**	**B**
13.	**B**	**14.**	**A**	**15.**	**E**	**16.**	**B**	**17.**	**B**	**18.**	**D**
19.	**B**	**20.**	**B**								

Solutions

1. A (792) — Numbers with 1, 2, 3, ... digits: 9 × 1 + 90 × 2 + 201 × 3 = 792. (There are 90 two-digit numbers: 10 to 99.) [2007 Q23 · one_to_round]

2. E (the piece with the 12) — The sums are: 1 = 1, 2 = 2, 3+4+5+6 = 18, 7 = 7, 8+9+10+11+12 = 50. [2002 Q11 · biggest_piece]

3. D (54) — The aquarium holds 4 × 4 × 4 = 64 cubes. There are already 10 inside (count column by column, also the hidden ones). 64 – 10 = 54. [2007 Q14 · how_many_more]

4. E — The empty area has 7 squares; all pieces do too, so look at the shape. Turn piece E and it fits exactly; each of the others has one square in a different place. [2003 Q18 · one_piece]

5. E (8) — Only the corner cubes have 3 painted faces, and a cube always has 8 corners, however big it is. [2007 Q17 · always_eight]

6. C (24) — 12 = 2 + 7 + 3; 21 = 11 + 7 + 3; 31 = 11 + 17 + 3; 35 = 18 + 7 + 10. No route gives 24. [2007 Q1 · impossible_total]

7. C (45) — The hole is 3 × 3 = 9 cubes in the front and goes 5 cubes back: 9 × 5 = 45. [2002 Q20 · fill_the_hole]

8. E — Turning or folding can only make the flag look narrower. Picture E reaches more than 4 squares away from the pole, which is wider than the flag. [2002 Q15 · too_wide]

9. B (10) — 4 m = 400 cm. 400 ÷ 40 = 10. [2003 Q10 · pieces]

10. E (3) — 1 + 3 × 8 = 25. The other digits do not work with 1 and 8. [2002 Q21 · hidden_card]

11. A (36) — The 6th diagram is 13 × 13 = 169 cells, with 6 × 6 = 36 black ones. [2007 Q21 · black_and_total]

12. B (Sofia plays the drum) — The only possible matching: Sofia – drum; Zoe – flute; Nicolas – harp; Chloe – xylophone; Lucas – violin. [2002 Q18 · which_true]

13. B (8679) — The numbers with all digits the same are 11111, 22222, ... The next one after 13543 is 22222: 22222 – 13543 = 8679. [2002 Q24 · same_digits]

14. A — A single cut appears four times, the same on the left and right and on the top and bottom. Only design A looks like that (notches in the middle of the left and right sides). [2007 Q24 · which_can]

15. E (213) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 531 – 189 = 342 digits are three-digit numbers: 342 ÷ 3 = 114, from 100 to 213. [2003 Q16 · three_digit_pages]

16. B (1) — Only piece 2 fits. Not: 1 (its flat part between the two features has the wrong length); 3 (it has a round dome where the line has two small bumps); 4 (its step is taller than in the line); 5 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]

17. B (15 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 40 ÷ 8 = 5, so the side is 3 × 5 = 15 cm. [2002 Q6 · strips_to_square]

18. D (4) — Starting black and ending white, the bars come in pairs: 11 black and 11 white. Wide: 11 – 4 = 7. Narrow: 11 – 7 = 4. [2003 Q17 · ends_differ]

19. B (75 g) — Swap each pineapple for 3 plums: 1 × 3 + 3 = 6 plums weigh 450 g. One plum: 450 ÷ 6 = 75 g. [2002 Q9 · grams]

20. B (January 1, 2017) — 13 minutes after 11:47 p.m. it is midnight, and a new day and a new year start. Peter was born 32 minutes after midnight, on January 1, 2017. So Basil is older, but they have different birthdays and were even born in different years. [2007 Q12 · new_year_twins]